SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – CODING CHALLENGE

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Problem 1: User Registration Validation using Regular Expressions

Problem Statement:
A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:
• Email Address
• Password
• Mobile Number

Validation Rules:
Email:
- Must start with an alphabet.
- May contain letters, digits, '.', '_' before '@'.
- Domain should contain only alphabets.
- Valid extensions: .com, .org, .edu, .net, .in

Password:
- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one digit
- At least one special character (@,#,$,%,&,!)

Mobile Number:
- Must contain exactly 10 digits.
- First digit should be between 6–9.

Expected Output:
Display:
- Valid Email / Invalid Email
- Strong Password / Weak Password
- Valid Mobile Number / Invalid Mobile Number

Write the complete Python program using Regular Expressions.

Problem 2: Deterministic Finite Automata (DFA) Simulator

Problem Statement:
Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:
• Accept the DFA description:
 - States
 - Input alphabet
 - Transition table
 - Initial state
 - Final state(s)
• Accept multiple input strings.
• Simulate state transitions.
• Display whether each string is Accepted or Rejected.

Example:
Language: Strings ending with "ab"

Input:
abaab

Output:
Transition Path:
q0 → q1 → q0 → q1 → q2
Accepted

Write the complete Python program.

Problem 3: Text Search Engine using Regular Expressions

Problem Statement:
Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:
• Word Search
• Prefix Search
• Suffix Search
• Date Search
• Phone Number Search
• Hashtag Search
• Mention (@username) Search

Example Input:

Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

User Menu:
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix

Expected Output:
Display all matching patterns based on user selection.

Write the complete Python program using Regular Expressions.

Submission Guidelines

1. Write well-commented Python programs.
2. Include sample input and output for each problem.
3. Submit the Python source code (.py) and this assignment document.
4. Ensure the programs execute without errors.